

AI Resume Analyzer and Job Recommendation System

Project Report

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Abstract

This report outlines the development of an NLP-powered application designed to evaluate technical resumes against industry job roles. The system extracts unstructured document data, normalizes technical terminology, and ranks candidate suitability using TF-IDF vectorization and cosine similarity. By integrating Google Gemini AI and a LangChain conversational coach, the application provides candidates with deterministic skill gap analyses, personalized 4-week learning roadmaps, and actionable career guidance, bridging the gap between student resumes and industry expectations.

System Architecture

The system follows a sequential pipeline, converting unstructured resume data into actionable career intelligence and generated roadmaps:

1. **Document Upload & Extraction:** Accepts PDF and DOCX files via a Streamlit interface, extracting raw text utilizing `pypdf` and `python-docx`
2. **Text Normalization:** Cleans punctuation and normalizes spacing while executing regex-based token protection to preserve critical technical terms (e.g., C++, .NET, React.js).
3. **Taxonomy-Based Skill Extraction:** Scans the normalized text against a controlled dictionary of technical skills grouped into domains, utilizing exact word-boundary regex `(\b)` to prevent false positive matches.
4. **Vectorization & Matching:** Computes 1D dense vectors for the candidate's skills and predefined job roles using `scikit-learn`'s TF-IDF vectorizer. It employs cosine similarity to generate an empirical percentage match score.
5. **Gap Analysis:** Deterministically calculates exact strengths (found skills) and gaps (missing skills) by comparing the candidate's extracted skill set against the target role's requirements.
6. **Roadmap Generation:** Integrates with the Google Gemini API (`gemini-3.5-flash-lite`) to synthesize the skill gaps into a structured, actionable 4-week learning plan.
7. **UI & Reporting:** Renders interactive Plotly charts, an LCEL-powered LangChain chat assistant, and exports the final analysis as a formatted PDF utilizing `fpdf`.

Key Modules and Implementation

Document Processing and Extraction

The `resume_parser.py` and `text_cleaner.py` modules handle the ETL phase. They ensure the robust ingestion of unstructured data and sanitize it for downstream NLP tasks. The

`skill_extractor.py` module acts as the taxonomy engine, rapidly identifying core competencies without the heavy overhead of neural embedding models, ensuring lightning-fast execution times.

Scoring Pipeline and AI Generation

The `job_matcher.py` acts as the core mathematical engine, utilizing sparse TF-IDF matrices to rank roles. To provide actionable value, `roadmap_generator.py` prompts the Gemini LLM with the exact missing skills to generate personalized curricula. Furthermore, `resume_coach.py` establishes a LangChain conversational agent equipped with the full context of the candidate's resume, match scores, and learning plan to answer specific career queries.

Conclusion

The AI Resume Analyzer successfully demonstrates a complete, end-to-end career guidance system. The integration of classical, lightweight NLP techniques (TF-IDF, Regex) for high-speed deterministic matching, combined with the reasoning and generation capabilities of modern LLMs (Gemini, LangChain), yields a highly dependable, low-latency tool that empowers candidates to optimize their professional profiles and address critical skill gaps.